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Comparison Between Linux and Windows Operating Systems

- ## Introduction:

Two of the most widely used operating systems today are Linux and Microsoft Windows which differ significantly in architecture and usage .

Linux is an open -source, Unix-like operating system originally developed by Linux Torvalds in 1991.Linux was originally designed as a clone of UNIX and is released under the copyleft HPL license. Linux was developed following the lack of a working kernel for GNU,a UNIX-compatible operating system made entirely of free software that had been in develompent since 1993 by the GNU Project. [12]

Microsoft Windows is a graphical operating system developed and marketed by Microsoft. It is the most popular desktop operating system in the world. It is grouped into families – Windows for personal computers, Windows server for servers, and Windows IoT for embedded systems. [11]

Both OS supports essential OS functionalties, some of which are process management, memory management, file systems, and synchronization mechanisms. However, they work differently for core OS problems like CPU scheduling,deadlocks, semaphores, and synchronization.[1]

➢ General Comparison:

Key features of Linux:

- Open-source nature – Distributors such as Ubuntu, Debian, Fedora offer full access to the source code giving users access to the entire codebase.
- Security and Stability – Known for extreme security providing better protection against malwares and viruses.
- Simple, Efficient and lightweight.
- Diverse Software Support – Users can install a quite large number of open-source softwares.

Key features of Windows:

- GUI – The interface is appealing and easy to use.
- Software compatibility – Supports a large amount of third-party applications that are not available on Linux.
- Gaming Performance – Offers better performance as it supports high-end GPU drivers.
- Hardware support – Works seamlessly with wide range of hardware devices
- regular updates – receives security patches and feature updates [10]

➢ CPU Scheduling algorithms:

Linux:

- Uses Completely Fair Scheduler:
  - ❖ It is a preemptive, priority-based algorithm with two separate priority ranges. One is the priority that is given to real-time proceses and the other is the priority that is assigned to a process based on the nice value of the process. The nice value ranges from -20 to 19. Smaller nice value indicates higher priorities.
  - ❖ Focuses on fair CPU time distribution.
  - ❖ Supports real time CPU scheduling (FIFO, Round Robin) [8]

Windows :

- Uses Priority-based preemptive scheduling.
  - ❖ Windows uses a preamptive, priority drive, round-robin scheduling approach, where each process or thread is assigned a priority.The CPU primarily executes the highest priority tasks, and lower

priority-tasks run when highest ones are waiting or blocked. [9]

- ❖ There are 32 priority levels :
  - ✓ Real time range (16-31)
  - ✓ Variable range (1-15)
  - ✓ Idle (0)
- ▪ Supports multilevel feedback queues.

➢Process Synchronisation:

Linux:

- Linux uses it to prevent race conditions
- Linux supports synchronisation between process, threads and kernel tasks
- Common synchronisation mechanisms include:

1) Mutex
2) Semaphores
3) Spinlocks
4) Conditional Variables [21]

Windows:

- Uses synchronization mechanisms to coordinate access to shared resources
- Common synchronisation tools include:
  1) Mutex
  2) Semaphores
  3) Critical Section
  4) Events
  5) Conditional Variables

- Windows API are provided by Win32 API
- Mostly used in GUI applications [22]

➢ Deadlock Handling:

Linux :

- Linux does not actively prevent deadlocks.
- It uses the Ostrich Algorithm, where it ignores the possibility of deadlock at the kernel level itself.
- Some tools are used:
    a) Lock Validator
    b) Out-of-Memory Kiler
    c) Wound/Wait Mutexes
- Avoidance is programmers responsibility. [4]

Windows:

- Deadlock Detection:
    - ❖ Wait Chain Traversal
    - ❖ Tools:
        a) Task Manager

b) Resource Monitoring
c) Windows Debugging Tools
d) Process Explorer

- Four necessary conditions for deadlocks:
  a) Mutual Exclusion
  b) Hold and Wait
  c) No preemption
  d) Circular wait [3]

➤ Semaphores:

Linux:

a) Linux support POSIX semaphores (sem_wait, sem_post)
b) Widely used in kernel and user space
c) Used in:
   - Process synchornization
   - Inter-process communication (IPC)
d) Gives low level control to the user.
e) More flexible
f) More complex [17]

Windows:

a) Windows uses Win32 API (CreateSemaphore, ReleaseSemaphore)
b) These are integrated with thread synchronization.
c) Implemented as kernel objects
d) Easier to use.
e) Less flexible. [18]

➢ Memory management:

Linux:

- Linux employs a virtual memory system,providing processes with illusion of having a private memory space.
- Virtual memory enables memory isolation,security and flexible allocation
- Paging - Both physical and virtual memory are divided into fixed-sized blocks called pages
- Kernel handles memory allocation through functions like kmalloc()
- Also TLB is maintained by CPU [15]

Windows:

- Uses virtual memory giving each process its own private address space
- Windows mainly uses paging
- Uses a page table to map logical address with physical address
- Uses a page file on disk as secondary memory when RAM becomes full
- Uses heap memory management
- Supports shared memory and DLL sharing
- Uses Address Space Layout Randomization for security against memory attacks [16]

➢Multithreading:

Linux:

- Linux Supports multithreading using POSIX Threads
- Each thread has its own stack, registres and program counter
- Linux kernel support kernel level threads
- Synchronisation is usually done using:
  1) Mutex

2) Semaphores
3) Conditional variables
4) Spinlocks[19]

Windows:

- Windows support multithreading via the Win32 API
- Threads under the same process share:
    a) Memory
    b) Handles
    c) Resources
- Each thread has its own stack, register set and execution state
- Windows too supports kernel level multithreading
- Synchronisation is done using:
    1) Mutex
    2) Semaphores
    3) Critical Section
    4) Events [20]

➢ Some interesting facts:

Linux:

- 90% of supercomputers run Linux.
- Android OS is based on Linux Kernel
- Used in:

1) NASA systems
2) Web servers
3) Banking infrastructure

Windows:

a) Known for:
   1) Huge software ecosystem
   2) Gaming dominance

WHY IS LINUX PREFFERED in Operating Systems?

Gives direct access to:

a) System calls
b) Process control
c) Memory management

- Strong security
- Low resource usage
- Inbuilt support for C, C++
- Built in compilers
- Bash scripting possible
- Free and open source
- Scalable

When and why is windows used?

- Easy to use GUI

- Good hardware support
- Best support for gaming
- GPU driver support
- Supports:
    - MS Office
    - Photoshop
- Strong support for GUI-based applications
- Beginner friendly
- Easy to understand the file management system.

- **Linux provides deeper functionalities into OS concepts due to its open architecture.**
- **Windows emphasizes user ease of access and application support.**